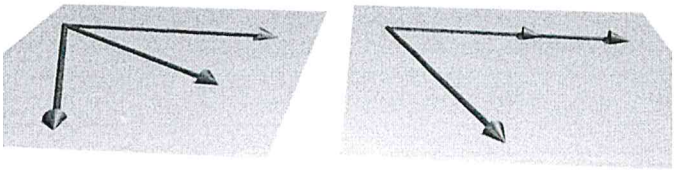
The Vector Equation of a Plane in R^3

Recall: If $\vec{w}$ is coplanar with $\vec{a}$ and $\vec{b}$, and $\vec{a}$ and $\vec{b}$ are not collinear, then we can express $\vec{w}$ as a LINEAR COMBINATION of $\vec{a}$ and $\vec{b}$.

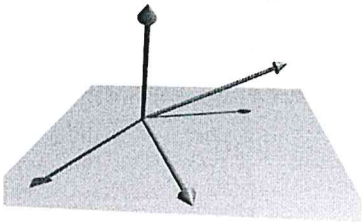
$\vec{w} = s\vec{a} + t\vec{b}$, $s, t \in \mathbb{R}$.

This is also called LINEARLY DEPENDENT. A set of vectors is said to be linearly dependent if one of the vectors in the set can be defined as a linear combination of the others.

In R^3, if two non-zero vectors are NOT COLLINEAR, they will be considered as LINEARLY INDEPENDENT and can form a BASIS for the space.



Linearly Dependent



Linearly Independent

Derive Vector Equation of a Plane in R^3

Let $\vec{a}$ and $\vec{b}$ be two non-collinear vectors (linearly independent) that lie in a plane.

Let $P_0(x_0, y_0, z_0)$ be a known point on the plane.

Let $P(x, y, z)$ be any point on the plane.

$\therefore \overrightarrow{P_0P}$ is a vector on the plane
$\therefore$ It is coplanar with $\vec{a}$ & $\vec{b}$
$\Rightarrow \therefore \overrightarrow{P_0P} = s\vec{a} + t\vec{b}$, $s, t \in \mathbb{R}$

$\therefore$ Now, $\overrightarrow{OP} = \overrightarrow{OP_0} + \overrightarrow{P_0P}$
$= \overrightarrow{OP_0} + (s\vec{a} + t\vec{b})$, $s, t \in \mathbb{R}$

Vector Equation of a Plane in R^3

$$\vec{r} = \vec{r_0} + s\vec{a} + t\vec{b} \quad , \; s, t \in \mathbb{R}$$

$\vec{r_0}$ = position vector of a known point in the plane.

$\vec{a}, \vec{b}$ are direction vectors for the plane.

$$(x, y, z) = (x_0, y_0, z_0) + s(a_1, a_2, a_3) + t(b_1, b_2, b_3)$$

Parametric Equations of a Plane in R^3

$$\left.\begin{aligned} x &= x_0 + sa_1 + tb_1 \\ y &= y_0 + sa_2 + tb_2 \\ z &= z_0 + sa_3 + tb_3 \end{aligned}\right\} \; s, t \in \mathbb{R}$$

* many ways to express equation of the plane depending on how we choose $\vec{r_0}$, $\vec{a}$ & $\vec{b}$

Example 1: Determine the vector and parametric equations of the plane that contains the points $A(1,0,-3)$, $B(2,-3,1)$ and $C(3,5,-3)$.

$\vec{AB} = (1,-3,4)$
$\vec{AC} = (2,5,0)$
} not collinear since not scalar multiples
∴ use as direction vectors $\vec{a}$ & $\vec{b}$.

could also use $\vec{OB}$, $\vec{OC}$

∴ $\vec{r} = (1,0,-3) + s(1,-3,4) + t(2,5,0)$, $s,t \in \mathbb{R}$

$x = 1 + s + 2t$
$y = -3s + 5t$
$z = -3 + 4s$
} $s,t \in \mathbb{R}$

Example 2: Determine the vector equation of a plane that contains the point $P(-4,3,6)$ and the line $L: \vec{r} = (1,-2,1) + t(2,-1,0), t \in \mathbb{R}$

Direction vectors: $\vec{a} = (2,-1,0)$ from given line.

$\vec{b} = (1,-2,1) - (-4,3,6)$ from 2 given points. (not collinear to $\vec{a}$)
$= (5,-5,-5)$
OR $(1,-1,-1)$

∴ $\vec{r} = (1,-2,1) + s(2,-1,0) + t(5,-5,-5)$, $s,t \in \mathbb{R}$.

OR $\vec{r} = (1,-2,1) + s(2,-1,0) + k(1,-1,-1)$, $s,k \in \mathbb{R}$

Example 3: Does $A(4,5,-3)$ lie in the plane $\vec{r} = (4,1,6) + s(3,-2,1) + t(-6,6,-1)$, $s,t \in \mathbb{R}$?

Solve for unique s,t.

$(4,5,-3) = (4,1,6) + s(3,-2,1) + t(-6,6,-1)$

① $4 = 4 + 3s - 6t$ ⇒ $0 = 3s - 6t$
② $5 = 1 - 2s + 6t$ ⇒ ④ $4 = -2s + 6t$
③ $-3 = 6 + s - t$

$4 = s$

Sub $s=4$ into ①.
$0 = 3(4) - 6t$
$0 = 12 - 6t$
$2 = t$

* but check in ⑤ for * uniqueness.

LS $= -3$. RS $= 6 + (4) - (2) = 8$.

LS ≠ RS.
∴ $(4,5,-3)$ does not lie in the plane.

Challenge: Determine the point at which the plane $\vec{r} = (4,1,6) + s(3,-2,1) + t(-6,6,-1)$, $s,t \in \mathbb{R}$ intersects the z-axis.

Hint: Let $P(0,0,z)$ ↳ solve for s,t.

$P(0,0,\frac{17}{6})$